CFD interpretation practical: trimmed two-condition airfoil design

Instructions and reference material

You have **30 minutes**. Work individually. No calculator, network access, AI tool, code, or notes are permitted. Write concise, evidence-based answers in your examination booklet.

Scenario

A 625 kg cargo UAV must carry its payload at cruise and then satisfy a high-Mach dash screen. All sections are 12%-thick. The design variables are maximum camber m and camber-crest location p/c; incidence is *trimmed* separately at each flight condition to supply the required lift.

Reference relations

Use $g = 10$ N/kg and load factor $n = 1.2$:

$$W = mg, \qquad L_{\text{req}} = nW, \qquad L = q_\infty S C_L, \qquad D = q_\infty S C_D.$$

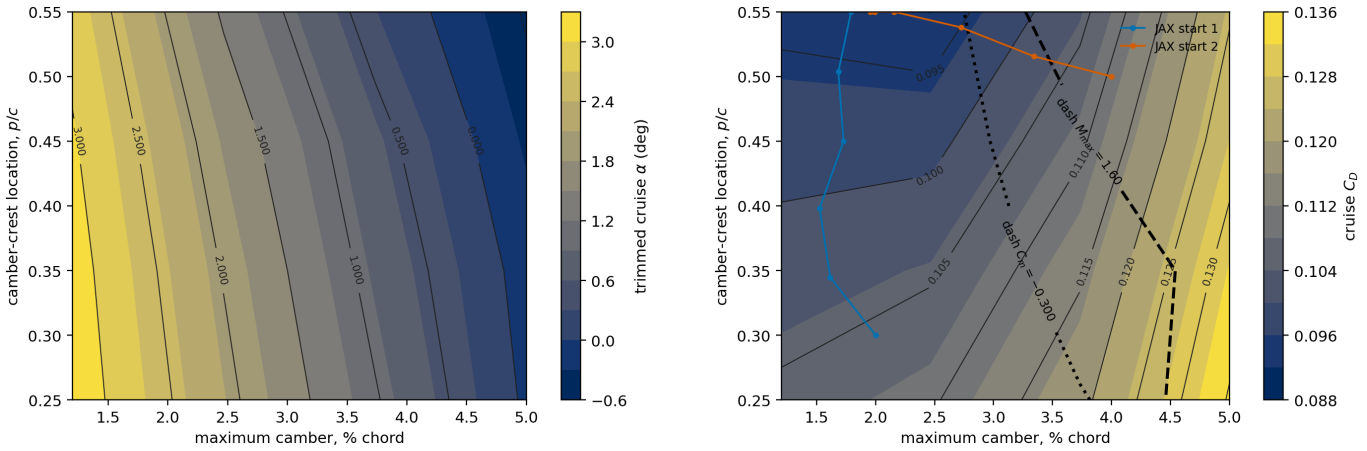
At cruise, $q_\infty S = 10.0$ kN and the trim target is $C_{L,\text{cruise}} = 0.75$. At dash, the trim target is $C_{L,\text{dash}} = 0.55$. The nominal dash limits are

$$M_{\text{max,dash}} \leq 1.60, \qquad C_{m,c/4,\text{dash}} \geq -0.300.$$

The JAX adjoint supplies the local derivatives used by the drag optimizer; it does not make the inviscid two-dimensional model a certification tool.

Model boundary. These calculations do not certify viscous drag, shock–boundary-layer interaction, separation, three-dimensional effects, structural loads, stability, or full aircraft performance.

1. Payload trim and JAX design maps [10 points]



Every map point is a separately trimmed solution. On the drag map, the black dashed contour is the dash Mach limit and the black dotted contour is the dash moment limit. Colored traces are two short JAX-adjoint gradient-descent histories from different starting geometries.

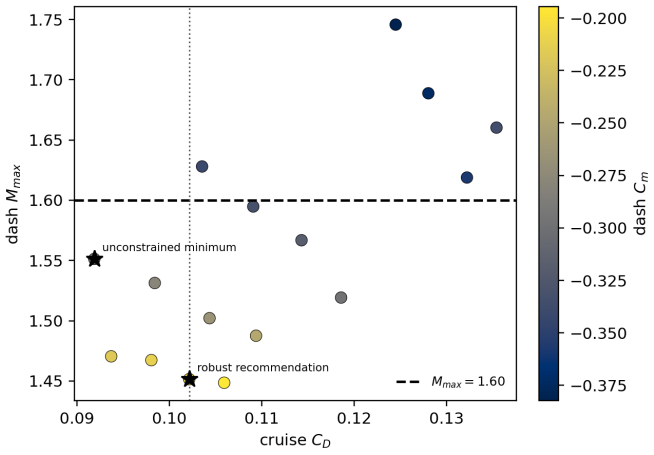
1. (2 points) Calculate L_{req} and verify $C_{L,\text{cruise}} = 0.75$ from the supplied cruise loading.

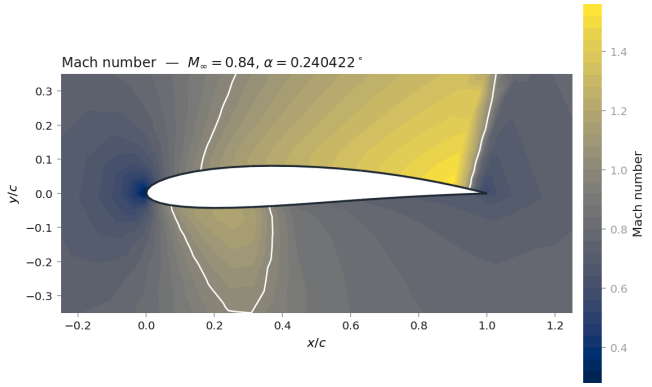
2. (2 points) Explain why choosing the largest camber or the largest incidence is not a valid design rule for this study.

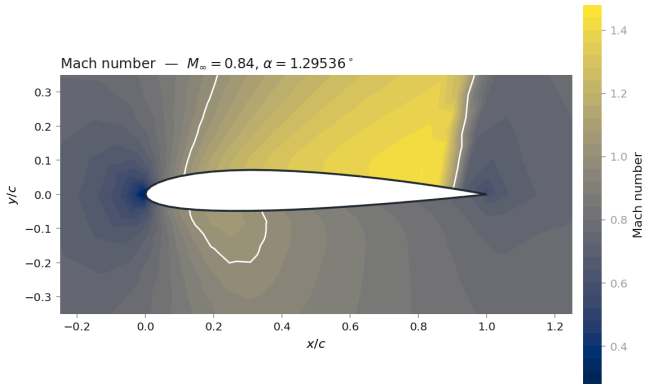
3. (3 points) On the drag map, identify the nominally feasible region. State which side of each constraint contour is acceptable; then determine whether the absolute minimum-drag point is feasible and explain why a minimum-drag point can nevertheless be a poor recommendation.

4. (3 points) Interpret one JAX optimization trace. Explain why it is useful that the adjoint supplies derivatives with respect to both m and p/c, even though a final engineering decision still requires constraint checks.

2. Robust recommendation and dash-flow evidence [10 points]







The stars identify the unconstrained grid minimum and the balanced recommendation. The recommendation is selected from nominally feasible grid points by maximizing its smaller normalized Mach/moment constraint margin, with drag used as a tie-breaker.

Candidate	m (% chord)	p/c	$C_{D,\text{cruise}}$	$M_{\text{max,dash}}$	$C_{m,\text{dash}}$
Unconstrained minimum	2.47	0.55	0.0919	1.551	-0.288
Balanced recommendation	1.20	0.35	0.1022	1.452	-0.204

5. (3 points) Compare the unconstrained drag minimum with the robust recommendation. State the trade-off that makes the robust choice an engineering recommendation rather than merely the nominal mathematical minimum.

6. (3 points) Use the dash trade-off plot and the two Mach fields to explain why a geometry with lower cruise drag can have less dash-flow margin, even when it still nominally passes the dash limits.

7. (2 points) If the robust design is used as a baseline, state one geometry or operating-condition change you would test next, and the physical risk it is intended to investigate.

8. (2 points) State the next numerical verification required before treating the drag ranking and constraint margins as reliable. Explain why this check matters particularly near a shock limit.

Solution key (instructor review)

Award credit for concise answers supported by the supplied artifacts. Populate any desired numerical tolerances from `candidate_summary.csv` after generating the figure set.

1. $W = 625(10) = 6250$ N and $L_{\text{req}} = 1.2(6250) = 7500$ N $= 7.5$ kN. Hence $C_{L,\text{cruise}} = L/(q_\infty S) = 7.5/10.0 = 0.75$.

2. Incidence is not independently selected: it is trimmed to meet lift. Higher camber changes the trimmed incidence, drag, dash peak Mach, and moment; a geometry must satisfy all constraints, not maximize one variable.

3. The feasible region lies on the low-$M_{\max}$ side of the dashed contour and the allowable-moment side of the dotted contour. Here the absolute drag minimum technically passes: $M_{\max} = 1.551 < 1.60$ and $C_m = -0.288 > -0.300$. It has only 0.049 Mach and 0.012 moment margin, however, so a small numerical or model change could make it fail.

4. The trace shows a gradient-based search in two geometry directions. JAX provides the local sensitivity of trimmed cruise drag to camber magnitude and crest location at the cost of an adjoint solve rather than independent full finite differences. Constraint values and model validity still must be checked separately.

5. The unconstrained minimum has $C_D = 0.0919$, $M_{\max} = 1.551$, and $C_m = -0.288$. The balanced recommendation accepts $C_D = 0.1022$ in exchange for $M_{\max} = 1.452$ and $C_m = -0.204$. Its Mach and moment margins are therefore much larger, protecting the recommendation against numerical/model uncertainty.

6. The lower-drag geometry produces a larger dash peak Mach and is visibly closer to a sharp upper-surface recompression. It remains nominally feasible, but its small margins make the conclusion fragile. The trade-off plot quantifies the margin; the Mach contours provide the flow evidence.

7. Accept a physically justified change, such as changing crest location while preserving cruise lift to control dash pressure recovery, or varying dash Mach/incidence to map shock onset. The answer must name the targeted risk: shock strength, wave drag, moment, or separation.

8. Perform a grid-refinement/mesh-convergence study, comparing trimmed incidence, C_D, $M_{\max}$, C_m, and shock location over successively finer meshes. Shock position and numerical drag are resolution-sensitive, so a small apparent margin can be numerical rather than physical.